

IP header

An **IP header** is a prefix to an IP packet that contains information about the IP version, length of the packet, source and destination IP addresses, etc. It consists of the following fields:

Version (4 bits)	Header length (4 bits)	Priority and Type of Service (8 bits)	Total length (16 bits)
Identification (16 bits)		Flags (3 bits)	Fragmented offset (13 bits)
Time to live (8 bits)	Protocol (8 bits)	Header checksum (16 bits)	
Source IP address (32 bits)			
Destination IP address (32 bits)			
Options (up to 32 bits)			

Here is a description of each field:

- **Version** – the version of the IP protocol. For IPv4, this field has a value of 4.
- **Header length** – the length of the header in 32-bit words. The minimum value is 20 bytes, and the maximum value is 60 bytes.
- **Priority and Type of Service** – specifies how the datagram should be handled. The first 3 bits are the priority bits.
- **Total length** – the length of the entire packet (header + data). The minimum length is 20 bytes, and the maximum is 65,535 bytes.
- **Identification** – used to differentiate fragmented packets from different datagrams.
- **Flags** – used to control or identify fragments.
- **Fragmented offset** – used for fragmentation and reassembly if the packet is too large to put in a frame.
- **Time to live** – limits a datagram's lifetime. If the packet doesn't get to its destination before the TTL expires, it is discarded.
- **Protocol** – defines the protocol used in the data portion of the IP datagram. For example, TCP is represented by the number 6 and UDP by 17.
- **Header checksum** – used for error-checking of the header. If a packet arrives at a router and the router calculates a different checksum than the one specified in this field, the packet will be discarded.
- **Source IP address** – the IP address of the host that sent the packet.
- **Destination IP address** – the IP address of the host that should receive the packet.
- **Options** – used for network testing, debugging, security, and more. This field is usually empty.

Consider the following IP header, captured with **Wireshark**:

```

Internet Protocol Version 4, Src: 192.168.5.45 (192.168.5.45), Dst: 91.198.174.192 (91.198.174.192)
  Version: 4
  Header Length: 20 bytes
  Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))
    0000 00.. = Differentiated Services Codepoint: Default (0x00)
    .... ..00 = Explicit Congestion Notification: Not-ECT (Not ECN-Capable Transport) (0x00)
  Total Length: 52
  Identification: 0x4116 (16662)
  Flags: 0x02 (Don't Fragment)
    0... .... = Reserved bit: Not set
    .1.. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
  Fragment offset: 0
  Time to live: 128
  Protocol: TCP (6)
  Header checksum: 0x0000 [validation disabled]
    [Good: False]
    [Bad: False]
  Source: 192.168.5.45 (192.168.5.45)
  Destination: 91.198.174.192 (91.198.174.192)
  [Source GeoIP: Unknown]
  [Destination GeoIP: Unknown]

```

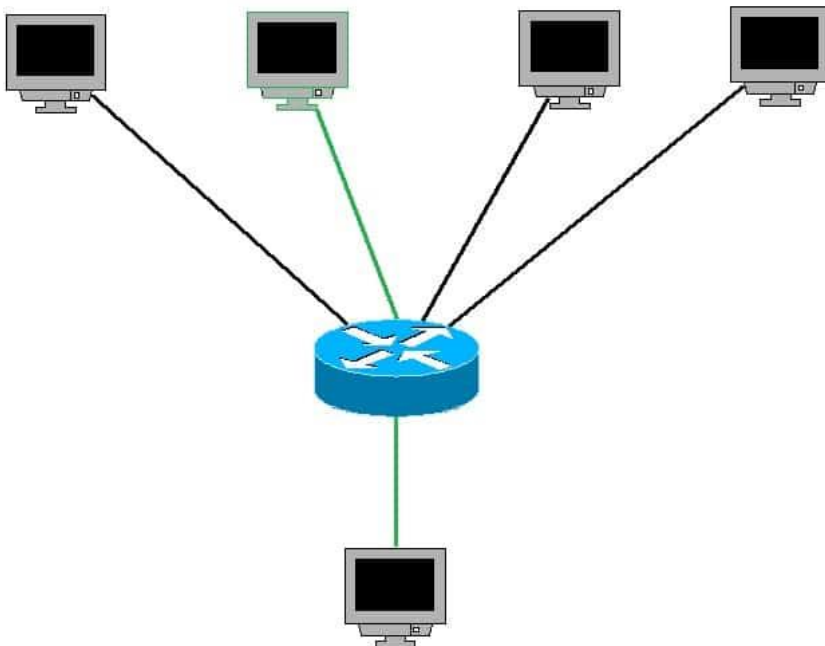
Notice the fields in the header: the IP version is IPv4, the header length is 20 bytes, the upper-level protocol used is TCP, the TTL value is set to 128, source and destination IP addresses are listed, etc.

Unicast, Multicast, and Broadcast Addresses

There are three types of Ethernet addresses:

1. Unicast Addresses

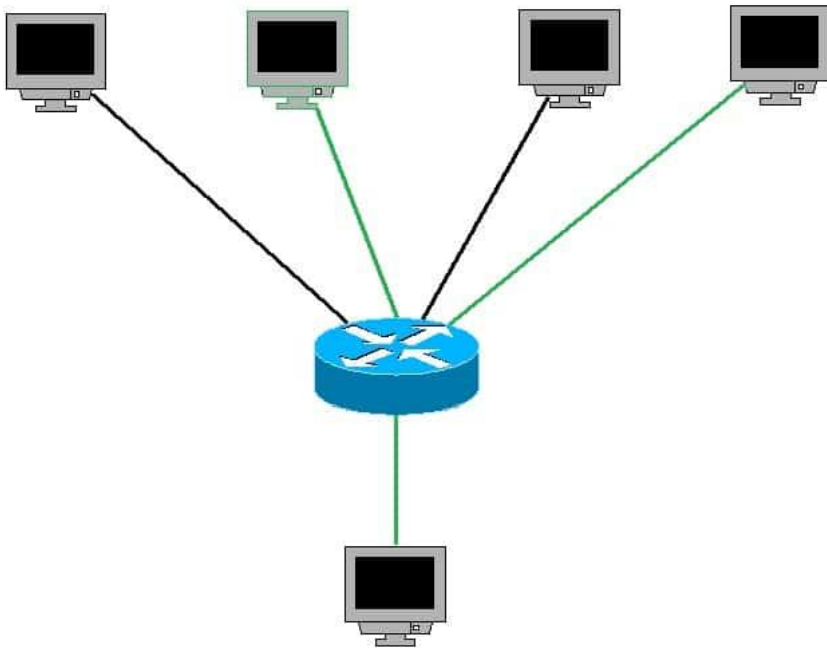
Unicast addresses represent a single LAN interface. A unicast frame will be sent to a specific device, not to a group of devices on the LAN:



The unicast address will have the value of the MAC address of the destination device.

2. Multicast Addresses

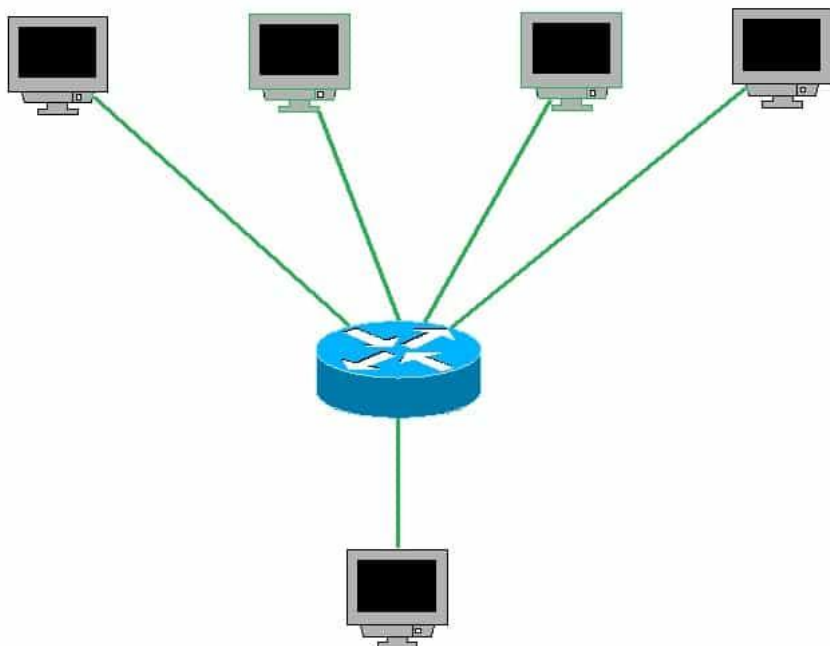
Multicast addresses represent a group of devices in a LAN. A frame sent to a multicast address will be forwarded to a group of devices on the LAN:



Multicast frames have a value of 1 in the least-significant bit of the first octet of the destination address. This helps a network switch to distinguish between unicast and multicast addresses. One example of an Ethernet multicast address would be **01:00:0C:CC:CC:CC**, which is the address used by [CDP \(Cisco Discovery Protocol\)](#).

3. Broadcast Addresses

Broadcast addresses represent all devices on the LAN. Frames sent to a broadcast address will be delivered to all devices on the LAN:



The broadcast address has the value of **FFFF.FFFF.FFFF** (all binary ones). The switch will flood broadcast frames out of all ports except the port that it was received on.

Types of IP Addresses

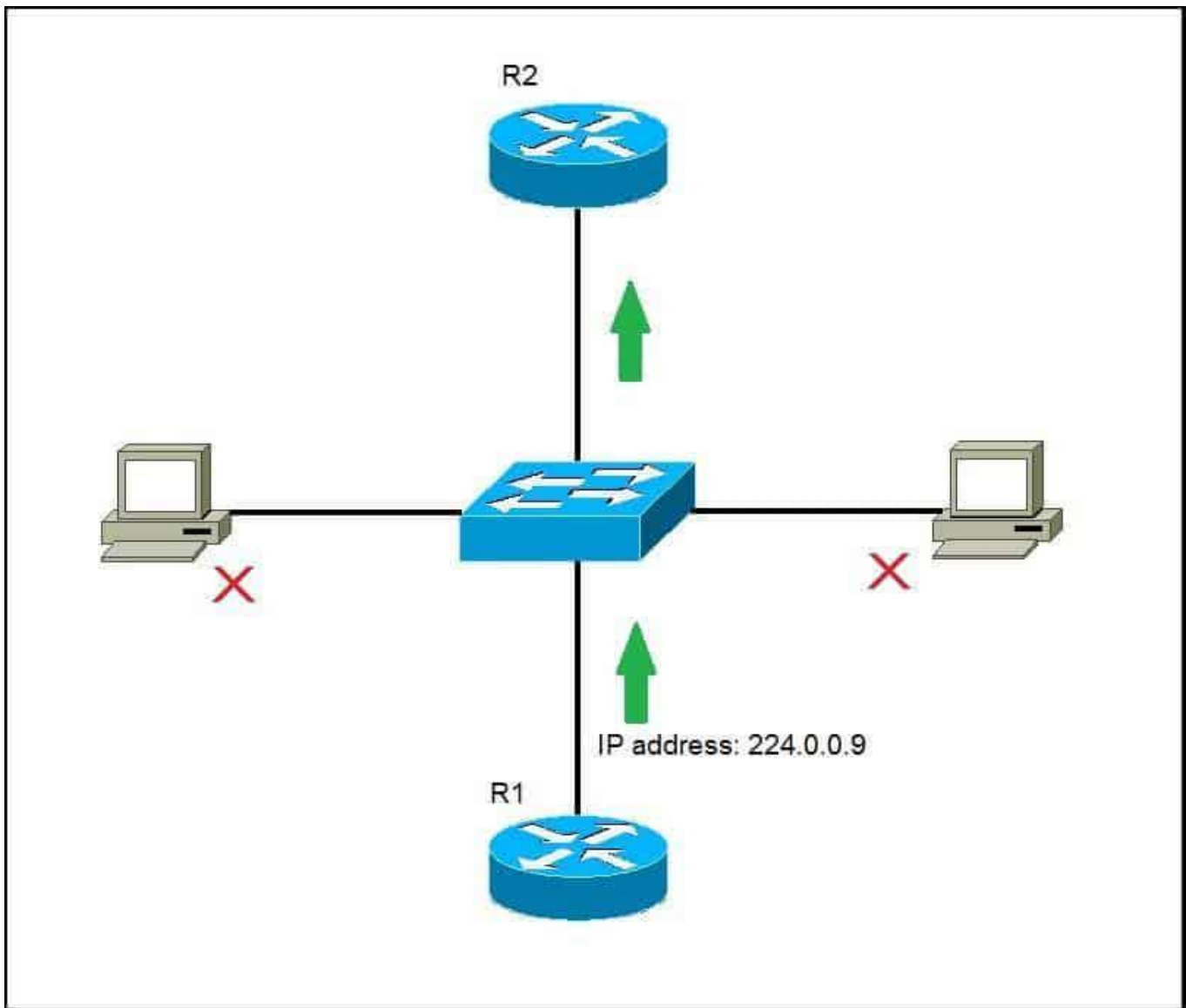
The IP addresses are divided into three different types, based on their operational characteristics:

1. unicast IP addresses – an address of a single interface. The IP addresses of this type are used for one-to-one communication. Unicast IP addresses are used to direct packets to a specific host. Here is an example:



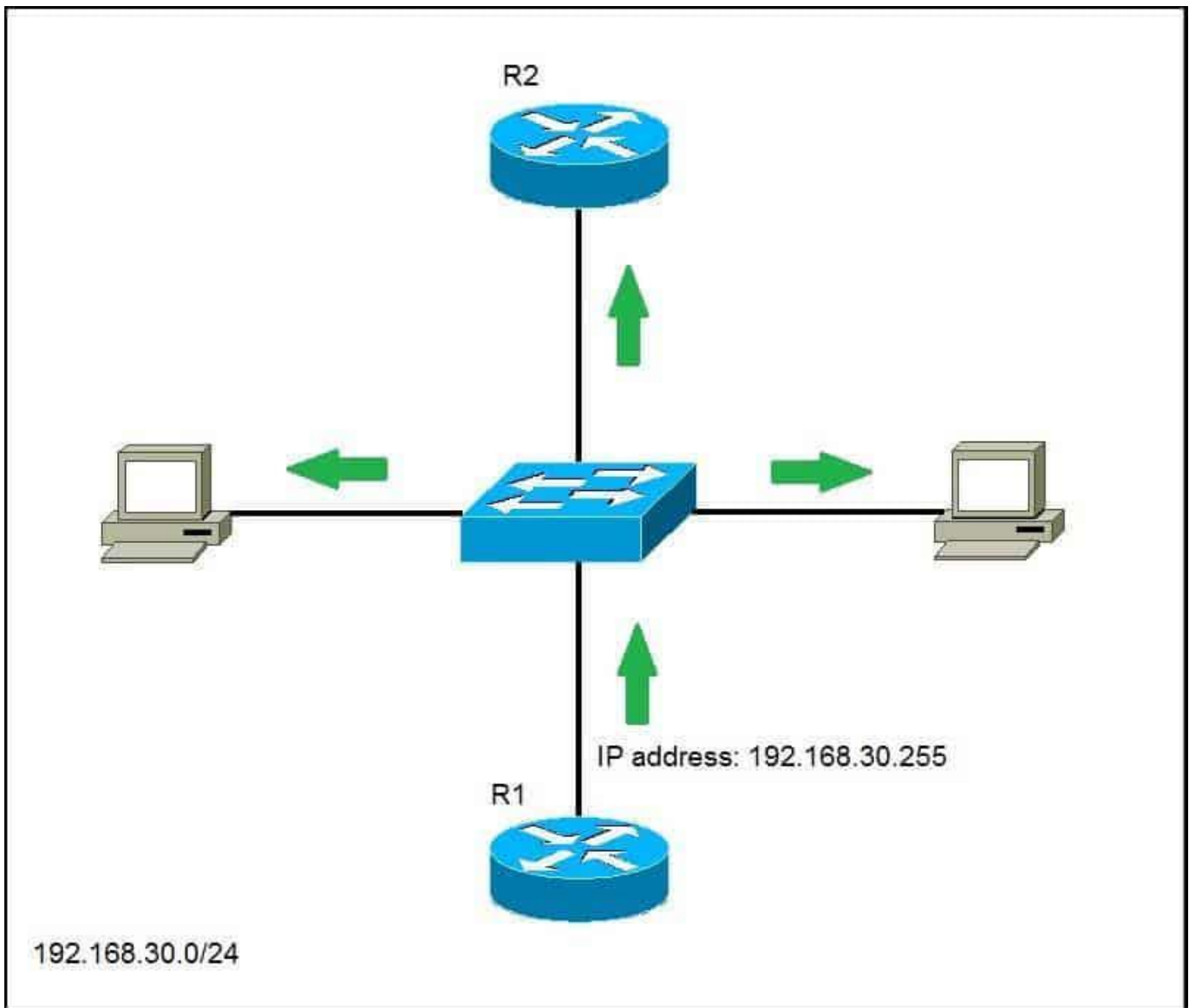
In the picture above you can see that the host wants to communicate with the server. It uses the (unicast) IP address of the server (**192.168.0.150**) to do so.

2. multicast IP addresses – used for one-to-many communication. Multicast messages are sent to IP multicast group addresses. Routers forward copies of the packet out to every interface that has hosts subscribed to that group address. Only the hosts that need to receive the message will process the packets. All other hosts on the LAN will discard them. Here is an example:



R1 has sent a multicast packet destined for **224.0.0.9**. This is an RIPv2 packet, and only routers on the network should read it. R2 will receive the packet and read it. All other hosts on the LAN will discard the packet.

3. broadcast IP addresses – used to send data to all possible destinations in the broadcast domain (the one-to-everybody communication). The broadcast address for a network has all host bits on. For example, for the network **192.168.30.0 255.255.255.0** the broadcast address would be **192.168.30.255***. Also, the IP address of all 1's (**255.255.255.255**) can be used for local broadcast. Here's an example:



R1 wants to communicate with all hosts on the network and has sent a broadcast packet to the broadcast IP address of 192.168.30.255. All hosts in the same broadcast domain will receive and process the packet.

**This is because the subnet mask of 255.255.255.0 means that the last octet in the IP address represents the host bits. And 8 one's written in decimal is 255.*

What is IPv4 Address and its Role in the Network?

IPv4 or Internet Protocol version 4, address is a 32-bit string of numbers separated by periods. It uniquely identifies a network interface in a device. IP is a part of the TCP/IP (Transmission Control Protocol/Internet Protocol) suite, where IP is the principal set of rules for communication on the Internet. An IP address is needed to be allocated on the devices, such as PCs, printers, servers, routers, switches, etc., to be able to communicate with each other in the network and out the Internet.

IPv4 Address Format

IPv4 addresses are expressed as a set of four numbers in decimal format, and each set is separated by a dot. Thus, the term 'dotted decimal format.' Each set is called an 'octet' because a set is

composed of 8 bits. The figure below shows the binary format of each octet in the 192.168.10.100 IP address:

Format	1 st Octet	2 nd Octet	3 rd Octet	4 th Octet
Dotted Decimal	192	168	10	100
Binary	1100 0000	1010 1000	0000 1010	0110 0100

A number in an octet can range from 0 to 255. Therefore, the full IPv4 address space goes from 0.0.0.0 to 255.255.255.255. The IPv4 address has two parts, the network part and the host part. A subnet mask is used to identify these parts.

Network Part

The network part of the IPv4 address is on the left-hand side of the IP address. It specifies the particular network to where the IPv4 address belongs. The network portion of the address also identifies the IP address class of the IPv4 address.

For example, we have the IPv4 address 192.168.10.100 and a /24 subnet mask. /24 simply means that the first 24 bits, starting from the left side, is the network portion of the IPv4 address. The 8 remaining bits of the 32 bits will be the host portion.

Format	1 st Octet	2 nd Octet	3 rd Octet	4 th Octet
Dotted Decimal	192	168	10	100
Binary	1100 0000	1010 1000	0000 1010	0110 0100



Host Part

The host portion of the IPv4 address uniquely identifies the device or the interface on your network. Hosts that have the same network portion can communicate with one another directly, without the need for the traffic to be routed.

IPv4 Address Allocation

The Internet Protocol address can be allocated to hosts or interfaces either manually or dynamically.

- **Static** – static IP address is set manually on the device. It is best practice to set static IP addresses on network devices, such as routers and switches, and on servers as well.

- **Dynamic** – dynamic IP address can be automatically allocated to a device via Dynamic Host Configuration Protocol (DHCP). Dynamic IP addresses are best to be used on end devices, such as PCs.

Types of IPv4 Addresses

We have two types of IP addresses, namely public IP addresses and private IP addresses.

- **Public IP address** – used to route Internet traffic. This is used on the Internet and is given out by Internet Service Providers (ISPs) to their customers.
- **Private IP address** – used in private networks for internal traffics within the LAN. Private addresses are not routable out the Internet.

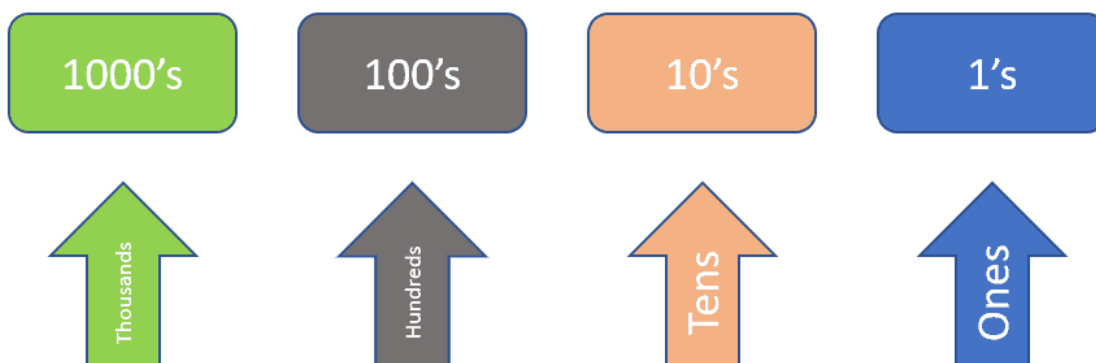
Converting the Decimal IP Address to Binary

IP addresses can be read in decimal or binary representation, but network devices only understand the binary value. Therefore, we must understand how to convert a decimal IP address to binary form.

Remember that IPv4 addresses are typically written in decimal digits, formatted as four 8-bit fields separated by periods. Each 8-bit field, called an octet, represents a byte of the IPv4 address.

Decimal Number

A decimal is a base 10 numbering system with which most of us are familiar. We use ten different numerals to represent the decimal numbers from zero to nine. The numerals are 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. Once we get to the number ten, there is no numeral to represent this value, so we go one place value up from ones to tens and so on and so forth.



Binary Number

Binary numbers are another number system mostly used to represent machine language. Unlike the decimal number, a binary number system only has two numerals to binary representations, which

are numbers 1 and 0. The placeholder in binary each has a value of, so Binary code is also known as the Base 2 Numbering System.

$$2^0 = 1$$

$$2^1 = 2$$

$$2^2 = 4$$

So since there are only two numerals that represent a binary digit, a usual example of it can be seen below.

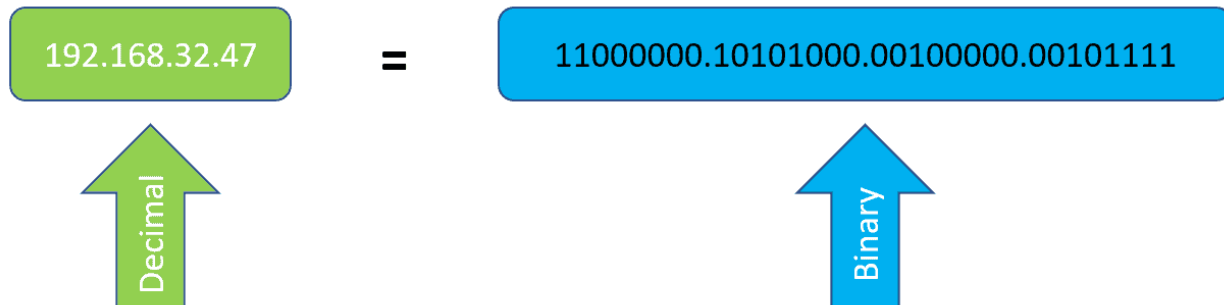
11100101

0001100

11110100

Converting Decimal to Binary IP Address

Below is an example of an IPv4 address in dotted decimal format and its corresponding binary format.



Understanding the IP address given in the example above, below are the corresponding equivalent of the decimal numbers to binary.

$$192 = 11000000$$

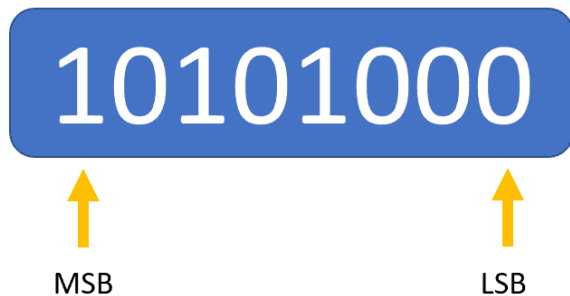
$$168 = 10101000$$

$$32 = 00100000$$

$$47 = 00101111$$

The first thing to do in converting decimal to binary system is to understand the corresponding decimal digits of the byte. As stated earlier, there are 8 bits in 1 byte, and every bit corresponds to specific digits based on the most significant bit (MSB) and the least significant bit (LSB).

8 bits = 1 Byte = Octet



Now we take the number 168, for example, and convert it to a binary number. To do so, we must do a simple addition method of values based on the table above to get the corresponding decimal number. Each 1's should be added, and 0's should be disregarded.

Now, subtract the decimal value from the MSB value of the octet and continue the process until you subtract the LSB value or your result is already zero. All the results with a positive value or zero will be counted as a 1, subtracted to the next value, and, if not, will be considered 0.

$$168 - 128 = 40 \text{ (1)}$$

$$40 - 64 = -24 \text{ (0)}$$

$$40 - 32 = 8 \text{ (1)}$$

$$8 - 16 = -8 \text{ (0)}$$

$$8 - 8 = 0 \text{ (1)}$$

$$0 - 4 = -4 \text{ (0)}$$

$$0 - 2 = -2 \text{ (0)}$$

$$0 - 1 = (0)$$

Once the difference is zero, the rest of the value up to the LSB shall be zero. Therefore the binary equivalent of **168** is **10101000**.

Subnet Mask Explained

An IP address is divided into two parts: network and host parts. For example, an IP class A address consists of 8 bits identifying the network and 24 bits identifying the host. This is because the default subnet mask for a class A IP address is 8 bits long. (or, written in dotted decimal notation, 255.0.0.0). What does it mean? Well, like an IP address, a subnet mask also consists of 32 bits. Computers use it to determine the network part and the host part of an address. The 1s in the subnet mask represent a network part, and the 0s a host part.

Computers work only with bits. The math used to determine a network range is binary AND.

INPUT 1	INPUT 2	OUTPUT
0	0	0
0	1	0
1	0	0
1	1	1

Let's say that we have the IP address of 10.0.0.1 with the default subnet mask of 8 bits (255.0.0.0). First, we need to convert the IP address to binary:

IP address: 10.0.0.1 = 00001010.00000000.00000000.00000001

Subnet mask 255.0.0.0 = 11111111.00000000.00000000.00000000

Computers then use the AND operation to determine the network number:

00001010.00000000.00000000.00000001 = 10.0.0.1

11111111.00000000.00000000.00000000 = 255.0.0.0

00001010.00000000.00000000.00000000 = 10.0.0.0

The computer can then determine the size of the network. Only IP addresses that begins with 10 will be in the same network. So, in this case, the range of addresses in this network is 10.0.0.0 – 10.255.255.255.

NOTE

A subnet mask must always be a series of 1s followed by a series of 0s.

Slash Notation

Aside from the dotted decimal format, we can also write the subnet mask in slash notation. It is a slash '/' then followed by the subnet mask bits. To determine the slash notation of the subnet mask, convert the dotted decimal format into binary, count the series of 1s, and add a slash on the start.

For example, we have the dotted decimal subnet mask of 255.0.0.0. In binary, it is 11111111.00000000.00000000.00000000. The number of succeeding 1s are 8, therefore the slash notation of 255.0.0.0 is /8.

Classes of IP addresses

TCP/IP defines five classes of IP addresses: class A, B, C, D, and E. Each class has a range of valid IP addresses. The value of the first octet determines the class. IP addresses from the first three classes (A, B and C) can be used for host addresses. The other two classes are used for other purposes – class D for multicast and class E for experimental purposes.

The system of IP address classes was developed for the purpose of Internet IP addresses assignment. The classes created were based on the network size. For example, for the small

number of networks with a very large number of hosts, the Class A was created. The Class C was created for numerous networks with small number of hosts.

Classes of IP addresses are:

Class	First octet value	Subnet mask
A	0-127	8
B	128-191	16
C	192-223	24
D	224-239	-
E	240-255	-

For the IP addresses from Class A, the first 8 bits (the first decimal number) represent the network part, while the remaining 24 bits represent the host part. For Class B, the first 16 bits (the first two numbers) represent the network part, while the remaining 16 bits represent the host part. For Class C, the first 24 bits represent the network part, while the remaining 8 bits represent the host part.

Consider the following IP addresses:

- **10.50.120.7** – because this is a Class A address, the first number (10) represents the network part, while the remainder of the address represents the host part (50.120.7). This means that, in order for devices to be on the same network, the first number of their IP addresses has to be the same for both devices. In this case, a device with the IP address of 10.47.8.4 is on the same network as the device with the IP address listed above. The device with the IP address 11.5.4.3 is not on the same network, because the first number of its IP address is different.
- **172.16.55.13** – because this is a Class B address, the first two numbers (172.16) represent the network part, while the remainder of the address represents the host part (55.13). A device with the IP address of 172.16.254.3 is on the same network, while a device with the IP address of 172.55.54.74 isn't.

NOTE

The system of network address ranges described here is generally bypassed today by use of the [Classless Inter-Domain Routing \(CIDR\)](#) addressing.

Special IP address ranges that are used for special purposes are:

- **0.0.0.0/8** – addresses used to communicate with the local network
- **127.0.0.0/8** – loopback addresses
- **169.254.0.0/16** – link-local addresses (APIPA)

<https://www.practicalnetworking.net/stand-alone/classful-cidr-flsm-vlsm/>

CIDR (Classless inter-domain routing)

CIDR (Classless inter-domain routing) is a method of public IP address assignment. It was introduced in 1993 by Internet Engineering Task Force with the following goals:

- to deal with the IPv4 address exhaustion problem
- to slow down the growth of routing tables on Internet routers

Before CIDR, public IP addresses were assigned based on the class boundaries:

- **Class A** – the classful subnet mask is /8. The number of possible IP addresses is 16,777,216 (2 to the power of 24).
- **Class B** – the classful subnet mask is /16. The number of addresses is 65,536
- **Class C** – the classful subnet mask is /24. Only 256 addresses are available.

Some organizations were known to have gotten an entire Class A public IP address (for example, IBM got all the addresses in the 9.0.0.0/8 range). Since these addresses can't be assigned to other companies, there was a shortage of available IPv4 addresses. Also, since IBM probably didn't need more than 16 million IP addresses, a lot of addresses were unused.

To combat this, the classful network scheme of allocating the IP address was abandoned. The new system was classless – a classful network was split into multiple smaller networks. For example, if a company needs 12 public IP addresses, it would get something like this: **190.5.4.16/28**.

The number of usable IP addresses can be calculated with the following formula:

2 to the power of host bits – 2

In the example above, the company got 14 usable IP addresses from the 190.5.4.16 – 190.5.4.31 range because there are 4 host bits and 2 to the power of 4 minus 2 is 14. The first and the last address are the network address and the broadcast address, respectively. All other addresses inside the range could be assigned to Internet hosts.

Create Subnets

There are a couple of ways to create subnets. In this article, we will subnet a class C address **192.168.0.0** that, by default, has **24** subnet bits and **8** host bits.

Before we start subnetting, we have to ask ourselves these two questions:

1. How many subnets do we need?

2^x = number of subnets. **x** is the number of 1s in the subnet mask. With 1 subnet bit, we can have 2^1 or 2 subnets. With 2 bits, 2^2 or 4 subnets, with 3 bits, 2^3 or 8 subnets, etc.

2. How many hosts per subnet do we need?

$2^y - 2$ = number of hosts per subnet. **y** is the number of 0s in the subnet mask.

Subnetting Example

An example will help you understand the subnetting concept. Let's say that we need to subnet a class C address **192.168.0.0/24**. We need two subnets with **50** hosts per subnet. Here is our calculation:

1. Since we need only two subnets, we need 2^1 subnet bits. In our case, this means that we will take one bit from the host part. Here is the calculation:

First, we have a class C address 192.168.0.0 with the subnet mask of 24. Let's convert them to binary:

192.168.0.0 = **11000000.10101000.00000000.00000000**
255.255.255.0 = **11111111.11111111.11111111.00000000**

We need to take a single zero from the host part of the subnet mask. Here is our new subnet mask:

255.255.255.128 = **11111111.11111111.11111111.10000000**

Remember, the ones in the subnet mask represent the network.

2. We need 50 hosts per subnet. Since we took one bit from the host part, we are left with seven bits for the hosts. Is it enough for 50 hosts? The formula to calculate the number of hosts is $2^y - 2$, with y representing the number of host bits. Since $2^7 - 2$ is 126, we have more than enough bits for our hosts.

3. Our network will look like this:

192.168.0.0/25 – the first subnet has the subnet number of **192.168.0.0**. The range of IP addresses in this subnet is **192.168.0.0 – 192.168.0.127**.

192.168.0.128/25 – the second subnet has the subnet number of **192.168.0.128**. The range of IP addresses in this subnet is **192.168.0.128 – 192.168.0.255**.

Understanding Variable Length Subnet Masks (VLSM)

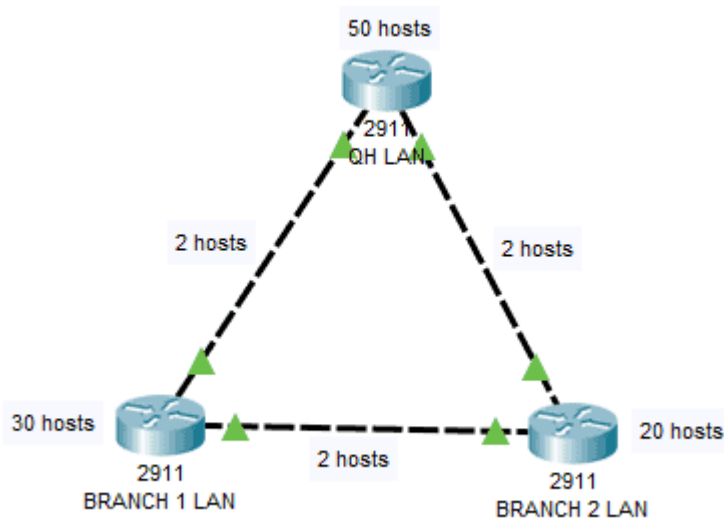
We have a limited number of private IPv4 addresses that can be used in every organization. As the Internet and most organizations are aggressively growing, we need a way to eliminate wasting IPv4 addresses. One of the ways that we can maximize the use of private IPv4 addresses in the organization is through subnetting.

The reason why we need subnetting is to efficiently distribute an IPv4 address with the least wastage and to create more networks with the smaller broadcast domains. To efficiently use subnetting, we can use Variable-Length Subnet Mask (VLSM).

With Variable-Length Subnet Mask (VLSM), we can allot the closest required number of IP addresses into a subnetwork in our LAN. We don't need to use a /23 subnet mask in all of our subnets, for example.

Steps to Implement Variable-Length Subnet Mask (VLSM)

We will use the below network topology as we go through the steps of the Variable-Length Subnet Masking (VLSM).



Step 1. Identify the host requirement. How many hosts or IP addresses are needed by the subnets in our LAN? We can arrange them from the highest host requirement to the lowest, as we will perform VLSM subnetting starting from the subnet with the highest host requirement. Don't forget to include the point-to-point (WAN) links as well.

- HQ LAN – 50 hosts
- BRANCH 1 – 30 hosts
- BRANCH 2 – 20 hosts
- WAN 1 (HQ to BRANCH 1) – 2 hosts
- WAN 2 (HQ to BRANCH 2) – 2 hosts
- WAN 3 (BRANCH 1 to BRANCH 2) – 2 hosts

The total host requirement for our network is 106 hosts, and we will perform VLSM subnetting on the HQ LAN subnetwork first.

Step 2. Determine the class of IP subnet. We need to determine the class of IP subnet that we will use based on the required number of hosts.

Class A has 16,777,216, Class B has 65,536, and Class C has 256 IP addresses. As per our network requirement, we need only 106 hosts, therefore we will use a Class C IP address space. In our example, we will use 192.168.10.0. It could also be that the organization bought an IP address space from the IP address authorities.

Step 3. Identify the host bits for every subnet. In our network topology example, HQ LAN has 50 hosts requirement, therefore we would have 6 host bits.

2⁶ host bits will give us 64 hosts, minus 2 for the network address and broadcast address, which is equal to 62 usable host addresses. It suffices our 50 hosts requirement for HQ LAN.

Step 4. Calculate the subnet mask. Identify the network bits and determine the subnet mask of the subnet. We can get the subnet mask by subtracting the host bits from 32 (the total IPv4 address bits). For HQ LAN, it's $32 - 6$ host bits, which is equal to a /26. The subnet mask for HQ LAN is /26 and its long format is 255.255.255.192

Step 5. Get the increment. To determine in which block of number should we go up, we can use the formula of $2^{\text{host bits}}$. For HQ LAN, it is 2^6 host bits, which will give us an increment of 64.

Step 6. Determine the network address, broadcast address, and IP address range. Starting from the base IP address, we will go up or increment in the value computed in Step 5.

For our network, we have a base IP address of 192.168.10.0. For HQ LAN, we will increment in a block of 64 as calculated in Step 5. Moreover, since it is in the Class C IP address space, as identified in Step 2, we will increment in the 4th octet.

That will be:

192.168.10.0 + 64 (Current subnet)

192.168.10.64 (Base IP address for the next subnet)

We determined that the network address for HQ LAN subnet is 192.168.10.0. The broadcast address will be 1 less than the next IP subnet. That's $192.168.10.64 - 1$, which is 192.168.10.63.

Finally, to get the HQ LAN usable IP address range, it is the IP address range in between the network address and the broadcast address, 192.168.10.1 to 192.168.10.62.

Completing the Variable-Length Subnet Mask Subnetting Process

Now, we are done with subnetting the HQ LAN. To fully implement VLSM, we need to do subnetting as well on the remaining LAN and WAN networks, which are BRANCH 1 LAN, BRANCH 2 LAN, WAN 1, WAN 2, and WAN 3.

The next subnet to be subnetted in VLSM will be the BRANCH 1 LAN as it has the next highest number of hosts. We will start with 192.168.10.64 as our network address as it is where we ended with our first IP subnet, HQ LAN.

Follow the steps we did on HQ LAN to perform VLSM subnetting on the remaining LAN and WAN subnets in our network diagram.

Below are the host bits, subnet mask, increment, network address, broadcast address, and usable IP address ranges of each subnet of the network topology we used in our example:

HQ LAN:

Number of Hosts – 50

Host Bits – 6 bits

Subnet Mask – /26 or 255.255.255.192

Increment – 64

Network Address – 192.168.10.0

Broadcast Address – 192.168.10.63

Usable IP Addresses – 192.168.10.1 to 192.168.10.62

BRANCH 1 LAN:

Number of Hosts – 30

Host Bits – 5 bits

Subnet Mask – /27 or 255.255.255.224

Increment – 32

Network Address – 192.168.10.64

Broadcast Address – 192.168.10.95

Usable IP Addresses – 192.168.10.65 to 192.168.10.94

BRANCH 2 LAN:

Number of Hosts – 20

Host Bits – 5 bits

Subnet Mask – /27 or 255.255.255.224

Increment – 32

Network Address – 192.168.10.96

Broadcast Address – 192.168.10.127

Usable IP Addresses – 192.168.10.97 to 192.168.10.126

WAN 1:

Number of Hosts – 2

Host Bits – 2 bits

Subnet Mask – /30 or 255.255.255.252

Increment – 4

Network Address – 192.168.10.128

Broadcast Address – 192.168.10.131

Usable IP Addresses – 192.168.10.129 to 192.168.10.130

WAN 2:

Number of Hosts – 2

Host Bits – 2 bits

Subnet Mask – /30 or 255.255.255.252

Increment – 4

Network Address – 192.168.10.132

Broadcast Address – 192.168.10.135

Usable IP Addresses – 192.168.10.133 to 192.168.10.134

WAN 3:

Number of Hosts – 2

Host Bits – 2 bits

Subnet Mask – /30 or 255.255.255.252

Increment – 4

Network Address – 192.168.10.136

Broadcast Address – 192.168.10.139

Usable IP Addresses – 192.168.10.137 to 192.168.10.138